

Sec 5.1. Classical gates

Def 5.1.

1) An elementary classical gate is a transformation given by

$$g: \{0,1\}^n \rightarrow \{0,1\}^n$$

$$(x_1, \dots, x_n) \mapsto g(x_1, \dots, x_n).$$

2) An extended classical gate is a transformation given by

$$g: \{0,1\}^n \rightarrow \{0,1\}^m$$

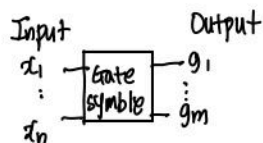
$$(x_1, \dots, x_n) \mapsto (g_1(x_1, \dots, x_n), \dots, g_m(x_1, \dots, x_n))$$

g_j : elementary

3) A classical gate g is reversible if it is bijective ($\Rightarrow$ invertible)

Description of classical gates:

1) Graphic symbol



2) Explicit formula

e.g. $u \oplus v := u + v \pmod{2}.$

Examples of classical gates

① NOT-gate : $\{0,1\} \rightarrow \{0,1\}$

$$x_1 \rightarrow \neg x_1 \quad , \quad \text{NOT}(x_1) = \neg x_1$$

$$\text{NOT}(0) = 1, \text{NOT}(1) = 0.$$

② AND-gate : $\{0,1\}^2 \rightarrow \{0,1\}$

$$x_1, x_2 \rightarrow x_1 x_2 \quad \text{AND}(x_1, x_2) = x_1 x_2$$

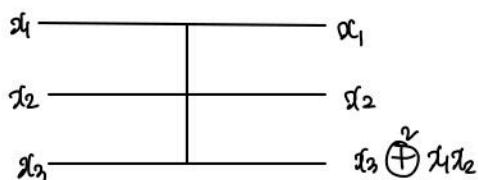
③ OR-gate : $\{0,1\}^2 \rightarrow \{0,1\}$

$$x_1, x_2 \rightarrow x_1 \vee x_2$$

④ XOR-gate : $\{0,1\}^2 \rightarrow \{0,1\}$

$$x_1, x_2 \rightarrow x_1 \oplus x_2$$

⑤ TOFFOLI-gate : $\{0,1\}^3 \rightarrow \{0,1\}^3$



$$\text{TOF}(x_1, x_2, x_3) = (x_1, x_2, x_3 \oplus x_1 x_2)$$

⑥

$$\text{ID}(x_1) = x_1$$

$$\text{FALSE}(x_1) = 0$$

$$\text{TRUE}(x_1) = 1$$

$$\text{COPY}^{(1)}(x_1) = (x_1, x_1)$$

Def 5.3.

i) $\mathcal{F}[g_1, \dots, g_k] :=$ the set of all gates which can be constructed by $g_1, \dots, g_k$ with the following construction rules.

$$\textcircled{i} \quad g_1, \dots, g_k \in \mathcal{F}[g_1, \dots, g_k]$$

$$\textcircled{ii} \quad n, \ell \in \mathbb{N}, y_1, \dots, y_\ell \in \{0, 1\}, \bar{y}_1, \dots, \bar{y}_\ell \in \{1, \dots, n+\ell\}.$$

$$\text{padding operation} \quad p_{y_1, \dots, y_\ell, \bar{y}_1, \dots, \bar{y}_\ell}^{(n)} \in \mathcal{F}[g_1, \dots, g_k]$$

$$: \{0, 1\}^n \longrightarrow \{0, 1\}^{n+\ell}$$

$$(x_1, \dots, x_n) \longmapsto (x_1, \dots, y_m, \dots, x_{n+\ell})$$

↓
 $\bar{y}_m$ -th slot.

$$\textcircled{iii} \quad n, \ell \in \mathbb{N}, \bar{y}_1, \dots, \bar{y}_\ell \in \{1, \dots, \ell\},$$

restriction

and / or

reordering

$$\text{operation} \quad r_{\bar{y}_1, \dots, \bar{y}_\ell}^{(n)} \in \mathcal{F}[g_1, \dots, g_k]$$

$$\{0, 1\}^n \longrightarrow \{0, 1\}^\ell$$

$$(x_1, \dots, x_n) \longmapsto (x_{\bar{y}_1}, \dots, x_{\bar{y}_\ell}).$$

$$\textcircled{iv} \quad h_1: \{0,1\}^n \rightarrow \{0,1\}^m, \quad h_2: \{0,1\}^l \rightarrow \{0,1\}^n \in \mathcal{H}[g_1, \dots, g_k]$$

$$\text{composition } h_1 \circ h_2 \in \mathcal{H}[g_1, \dots, g_k]$$

$$\textcircled{v} \quad h: \{0,1\}^n \rightarrow \{0,1\}^m, \quad k: \{0,1\}^p \rightarrow \{0,1\}^q \in \mathcal{H}[g_1, \dots, g_k]$$

$$\text{Cartesian } h \times k \in \mathcal{H}[g_1, \dots, g_k]$$

$$: \{0,1\}^{n+p} \rightarrow \{0,1\}^{m+q}$$

$$(x_1, \dots, x_n, x_{n+1}, \dots, x_{n+p}) \mapsto (h_1(x_1, \dots, x_n), \dots, h_m(x_1, \dots, x_n), k_1(x_{n+1}, \dots, x_{n+p}), \dots, k_q(x_{n+1}, \dots, x_{n+p}))$$

2) A set of classical gates $\mathcal{G} := \{g_1, \dots, g_J\}$ is universal

$$\text{iff } \forall g, \quad g \in \mathcal{H}[g_1, \dots, g_J]$$

3) When many of gates are connected, we call the resulting arrays "classical logical circuits".

Lemma 4. $h_1, \dots, h_L, g_1, \dots, g_k$

If $h_1, \dots, h_L \in \mathcal{H}[g_1, \dots, g_k]$, then $\mathcal{H}[h_1, \dots, h_L] \subset \mathcal{H}[g_1, \dots, g_k]$.

$$\textcircled{*} \quad \mathcal{H}[\mathcal{H}[g_1, \dots, g_k]] \subset \mathcal{H}[g_1, \dots, g_k].$$

$$\begin{aligned} & (\text{ID} \times \text{ID} \times \text{XOR}) \circ (\text{ID} \times \text{ID} \times \text{AND} \times \text{ID}) \circ \overset{\{5\}}{r_{[1,3,2,4,5]}} \circ \underbrace{(\text{COPY} \times \text{COPY} \times \text{ID})}_{//} (x_1, x_2, x_3) \\ & \quad // \\ & \quad (x_1, x_1, x_2, x_2, x_3) \\ & \quad // \\ & \quad (x_1, x_2, x_1, x_2, x_3) \\ & \quad // \\ & \quad (x_1, x_2, x_1 x_2, x_3) \\ & \quad // \\ & \quad (x_1, x_2, x_1 x_2 \oplus x_3) \\ & \quad // \\ & \quad \text{TOF}(x_1, x_2, x_3) \end{aligned}$$

②

Proof $\Rightarrow$

① $ID, FALSE, TRUE, NOT \in \mathcal{H}[TOF]$

$$\Downarrow$$

$$\text{COPY}^{(n)}(x_1, \dots, x_n) = (x_1, \dots, x_n, x_1, \dots, x_n)$$

$$\textcircled{3} \quad f = \text{XOR} \circ (\text{AND} \times \text{AND}) \circ (\underbrace{g_0 \times \text{NOT}}_{\text{AND}} \times (\underbrace{g_1 \times \text{ID}}_{\text{AND}})) \circ \text{COPY}^{(n)},$$

$$: \{0,1\}^{n+1} \rightarrow \{0,1\}$$

$\Rightarrow$ Induction assumption on $g_0, g_1 \Rightarrow f \in \mathcal{H}[\text{TOF}]$

(Invertibility) $\text{TOF} \circ \text{TOF} = \text{Id}$.



Sec 5.2. Quantum gates.

Def 5.1. A quantum n -gate is a unitary operator $U: \mathbb{H}^{\otimes n} \rightarrow \mathbb{H}^{\otimes n}$.

⊗ $n=1$, unary quantum gate

$n=2$, binary "

* U : completely determined by its action on a basis
& represented by a unitary matrix in that basis.

$$|x\rangle^n := |x_{n-1}, \dots, x_0\rangle$$

§1. Examples of Unary quantum gate:

unitary operator $V: \mathbb{H} \rightarrow \mathbb{H}$

unitary 2x2 matrix $V = \begin{bmatrix} v_{00} & v_{01} \\ v_{10} & v_{11} \end{bmatrix}$, $\{ |0\rangle, |1\rangle \}$

Name	Operator	Matrix in basis $\{ 0\rangle, 1\rangle\}$	symbol
Identity	1	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	
Phase-multiplication	$M(\alpha) := e^{i\alpha} 1$	$\begin{bmatrix} e^{i\alpha} & 0 \\ 0 & e^{i\alpha} \end{bmatrix}$	
Phase-shift	$P(\alpha) := 0\rangle\langle 0 + e^{i\alpha} 1\rangle\langle 1 $	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{bmatrix}$	
Pauli-X <i>quantum NOT</i>	$X := \sigma_x$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	
Pauli-Y	$Y := \sigma_y$	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	
Pauli-Z	$Z := \sigma_z$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	
HADAMARD	$H := \frac{1}{\sqrt{2}}(\sigma_y + \sigma_z)$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	
Spin-rotation by angle α around $\hat{n}$	$D_n(\alpha)$	$\begin{bmatrix} \cos \frac{\alpha}{2} - i \sin \frac{\alpha}{2} n_z & -i \sin \frac{\alpha}{2} (n_x - i n_y) \\ -i \sin \frac{\alpha}{2} (n_x + i n_y) & \cos \frac{\alpha}{2} + i \sin \frac{\alpha}{2} n_z \end{bmatrix}$	
Arbitrary unary gate	$V : \text{unitary}$	$\begin{bmatrix} v_{00} & v_{01} \\ v_{10} & v_{11} \end{bmatrix}$	

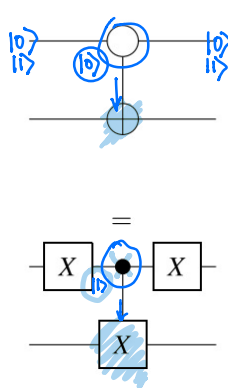
$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle.$$

§2. Binary quantum gates.

unitary operator $U \in \mathcal{U}(\mathbb{H}^{\otimes 2})$

Name	Symbol	Operator	Matrix in computational basis $\{ 0\rangle^2 = 00\rangle, 1\rangle^2 = 01\rangle$ $ 2\rangle^2 = 10\rangle, 3\rangle^2 = 11\rangle\}$
Controlled NOT (C-NOT)		$\Lambda^1(X)$ $=$ $ 0\rangle\langle 0 \otimes \mathbf{1}$ $+$ $ 1\rangle\langle 1 \otimes X$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$
C-NOT with control in 2. qbit		$\Lambda_1(X)$ $=$ $\mathbf{1} \otimes 0\rangle\langle 0 $ $+$ $X \otimes 1\rangle\langle 1 $ $=$ $H^{\otimes 2} \Lambda^1(X) H^{\otimes 2}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$
Controlled phase-multiplication		$\Lambda^1(M(\alpha))$ $=$ $ 0\rangle\langle 0 \otimes \mathbf{1}$ $+$ $ 1\rangle\langle 1 \otimes e^{i\alpha} \mathbf{1}$ $=$ $P(\alpha) \otimes \mathbf{1}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{i\alpha} & 0 \\ 0 & 0 & 0 & e^{i\alpha} \end{pmatrix}$

C-NOT
controlled
with $|0\rangle$

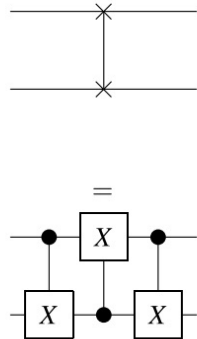


$$\Lambda^{[0]}(X) = |1\rangle\langle 1| \otimes \mathbf{1} + |0\rangle\langle 0| \otimes X$$

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$(X \otimes \mathbf{1}) \Lambda^1(X) (X \otimes \mathbf{1})$$

Swap

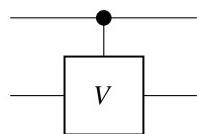


$$S = |00\rangle\langle 00| + |11\rangle\langle 11| + |01\rangle\langle 10| + |10\rangle\langle 01|$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\Lambda^1(X) \Lambda_1(X) \Lambda^1(X)$$

Controlled
 V



$$\Lambda^1(V) = |0\rangle\langle 0| \otimes \mathbf{1} + |1\rangle\langle 1| \otimes V$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & v_{00} & v_{01} \\ 0 & 0 & v_{10} & v_{11} \end{pmatrix}$$

$$(P(\alpha) \otimes A) \Lambda^1(X) (\mathbf{1} \otimes B) \Lambda^1(X) (\mathbf{1} \otimes C)$$

mit $V = M(\alpha) A X B X C$
und $\mathbf{1} = A B C$

§3. General quantum n -gates

Def 5.8. For $U_j \in \mathcal{U}(\mathbb{H}^{\otimes n_j})$, $j \in \{1, \dots, k\}$.

1) $\mathcal{F}[U_1, \dots, U_k] :=$ the set of all gates which can be constructed by $U_1, \dots, U_k$ with the following construction rules.

$$\textcircled{I} \quad U_1, \dots, U_k \in \mathcal{F}[U_1, \dots, U_k].$$

$$\textcircled{II} \quad n \in \mathbb{N}, \quad 1^{\otimes n} \in \mathcal{F}[U_1, \dots, U_k]$$

$$\textcircled{III} \quad V_1, V_2 \in \mathcal{U}(\mathbb{H}^{\otimes n}), \quad \text{if } V_1, V_2 \in \mathcal{F}[U_1, \dots, U_k], \quad V_1 V_2 \in \mathcal{F}[U_1, \dots, U_k]$$

$$\textcircled{IV} \quad V_1 \in \mathcal{U}(\mathbb{H}^{\otimes n_1}), \quad V_2 \in \mathcal{U}(\mathbb{H}^{\otimes n_2}),$$

$$\text{if } V_1, V_2 \in \mathcal{F}[U_1, \dots, U_k], \quad \text{then } V_1 \otimes V_2 \in \mathcal{F}[U_1, \dots, U_k]$$

2) A set of quantum gates $\mathcal{U} = \{U_1, \dots, U_k\}$ is universal

if for any gate V , $V \in \mathcal{F}[U_1, \dots, U_k]$, $U_1, \dots, U_k \in \mathcal{U}$.

Lemma 9. $V_1, \dots, V_L, U_1, \dots, U_k$

if $V_1, \dots, V_L \in \mathcal{F}[U_1, \dots, U_k]$, then $\mathcal{F}[V_1, \dots, V_L] \subset \mathcal{F}[U_1, \dots, U_k]$.

$$\textcircled{*} \quad \mathcal{F}[\mathcal{F}[U_1, \dots, U_k]] \subset \mathcal{F}[U_1, \dots, U_k].$$

Exer 5.59 & Def 5.10 $n, n_a, n_b \in \mathbb{N}_0$ with $n_a + n_b = n$ &

let $|a\rangle \in {}^1H^{\otimes n_a}$, $|b\rangle \in {}^1H^{\otimes n_b}$ be vectors in the respective computational basis. (eg. $|a\rangle = |0\rangle^{n_a}, \dots, |2^{n_a}-1\rangle^{n_a}$).

For any $V \in \mathcal{U}(H)$,

$$1^{\otimes (n+1)} + |a\rangle\langle a| \otimes (V-1) \otimes |b\rangle\langle b| \in \mathcal{U}({}^1H^{\otimes (n+1)}).$$

$$\Downarrow$$

$$\Delta_{|a\rangle, |b\rangle}^{(a,b)}(V) : (|a\rangle, |b\rangle)\text{-controlled } V$$

$$\textcircled{1} |a\rangle = |2^{n_a}-1\rangle, |b\rangle = |2^{n_b}-1\rangle \quad ; \quad \Delta_{n_b}^{n_a}(V).$$

$$\textcircled{2} n = n_a \Rightarrow a = 2^n - 1 \quad ; \quad \Delta^n(V)$$

$$n = n_b \Rightarrow b = 2^n - 1 \quad ; \quad \Delta_n(V).$$

$$\textcircled{3} n = 0 \quad ; \quad \Delta^0(V) := V =: \Delta_0(V)$$

Exer 5.60. $V \in \mathcal{U}({}^1H)$, $\alpha \in \mathbb{R}$,

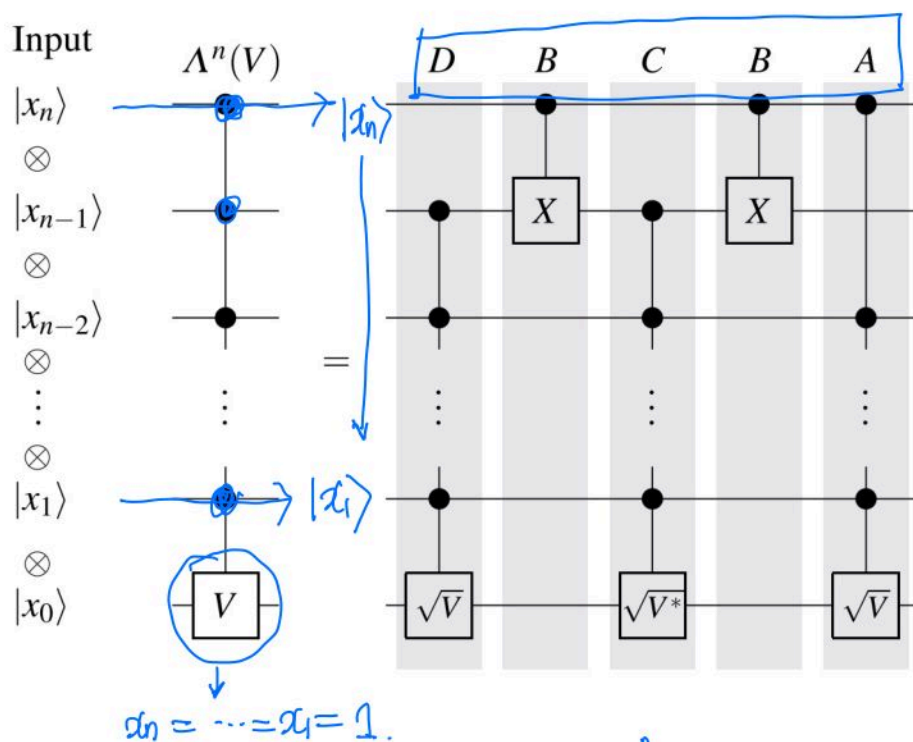
$$\left. \begin{aligned} \Delta^1(V) &= |0\rangle\langle 0| \otimes 1 + |1\rangle\langle 1| \otimes V. \\ \Delta_1(x) &= H^{\otimes 2} \Delta^1(x) H^{\otimes 2} \\ \Delta^1(M(\alpha)) &= P(\alpha) \otimes 1. \end{aligned} \right]$$

Thm 5.11 & Lem 5.12. $V \in \mathcal{U}(\mathcal{H})$ & $n \in \mathbb{N}_0$.

$$\Lambda_1(V), \Lambda^n(V) \in \mathcal{H}[M, \mathcal{P}_q, \mathcal{P}_\infty, \Lambda^1]$$

$$(*)_{n=0} \quad V \in \mathcal{H}[M, \mathcal{P}_q, \mathcal{P}_\infty].$$

$$|x\rangle^{n+1} = |x_n \dots x_1 x_0\rangle$$



$$\Lambda^n(V) = |x_n \dots x_1\rangle \otimes V^{\left(\prod_{j=1}^n x_j\right)} |x_0\rangle$$

$$\Lambda^n(V) = DBCBA, \quad D, C, A \in \mathcal{H}[\Lambda^{n+1}(\sqrt{V}), \Lambda^{n+1}(\sqrt{V^*})]$$

$$B \in [\Lambda^1(x)]$$

$$S (= |00\rangle\langle 00| + |11\rangle\langle 11| + |01\rangle\langle 10| + |10\rangle\langle 01| = \Lambda'_1(x) \Lambda_1(x) \Lambda'_1(x))$$

Def 5.13. $n \in \mathbb{N}$, j, k with $k < j \leq n-1$,

$$\begin{aligned} \textcircled{1} S_{jk}^{(n)} &:= 1^{\otimes n-j} \otimes |0\rangle\langle 0| \otimes 1^{\otimes j-k-1} \otimes |0\rangle\langle 0| \otimes 1^{\otimes k} \\ &\quad + 1^{\otimes n-j} \otimes |1\rangle\langle 1| \otimes 1^{\otimes j-k-1} \otimes |1\rangle\langle 1| \otimes 1^{\otimes k} \\ &\quad + 1^{\otimes n-j} \otimes |0\rangle\langle 1| \otimes 1^{\otimes j-k-1} \otimes |1\rangle\langle 0| \otimes 1^{\otimes k} \\ &\quad + 1^{\otimes n-j} \otimes |1\rangle\langle 0| \otimes 1^{\otimes j-k-1} \otimes |0\rangle\langle 1| \otimes 1^{\otimes k} \end{aligned}$$

$$S_{jj}^{(n)} := 1^{\otimes n}$$

$$\textcircled{2} S^{(n)} := \prod_{j=0}^{\lfloor \frac{n}{2} \rfloor - 1} S_{n-j, j}^{(n)} : \text{the global swap on } \mathbb{H}^{\otimes n}$$

Exer 5.16. For $\bigotimes_{\ell=n+1}^0 |\psi_\ell\rangle \in \mathbb{H}^{\otimes n}$

$$S_{jk}^{(n)} \bigotimes_{\ell=n+1}^0 |\psi_\ell\rangle = |\psi_{n+1} \dots \psi_k \psi_j \psi_1\rangle$$

$$\begin{aligned} S^{(n)} \bigotimes_{\ell=n+1}^0 |\psi_\ell\rangle &= \prod_{j=0}^{\lfloor \frac{n}{2} \rfloor - 1} S_{n-j, j}^{(n)} \bigotimes_{\ell=n+1}^0 |\psi_\ell\rangle \\ &= S_{n,0}^{(n)} S_{n-1,1}^{(n)} \dots S_{\lfloor \frac{n}{2} \rfloor - 1, \lfloor \frac{n}{2} \rfloor}^{(n)} |\psi_{n+1} \dots \psi_0\rangle \\ &= |\psi_0 \psi_1 \psi_2 \dots \psi_{n+1}\rangle \end{aligned}$$

Lemma 5.15. $V \in \mathcal{U}(\mathcal{H})$, $n_a, n_b \in \mathbb{N}_0$

$$\begin{aligned}\Delta_{n_b}^{n_a}(V) &= S_{n_b 0}^{(n_a+n_b+1)} \Delta^{n_a+n_b}(V) S_{n_b 0}^{(n_a+n_b+1)} \\ &\in \mathfrak{F}[\Delta^1(x), \Delta_1(x), \Delta^{n_a+n_b}(V)].\end{aligned}$$

$$\Delta_{|b\rangle}^{|a\rangle}(V)$$

Def 5.16. $A \in L(\mathcal{H})$, $|b\rangle :=$ the computational basis of $\mathcal{H}^{\otimes n}$
 $\parallel$
 $|b_{n-1} \dots b_1 b_0\rangle$

$$\Rightarrow A^{\otimes |b\rangle} := A^{b_{n-1}} \otimes \dots \otimes A^{b_0}$$

$$| \neg b \rangle := | \neg b_{n-1} \dots \neg b_0 \rangle = X | b_{n-1} \rangle \otimes \dots \otimes X | b_0 \rangle.$$

Lemma 5.17.

$$\begin{aligned}\Delta_{|b\rangle}^{|a\rangle}(V) &= [X^{\otimes | \neg a \rangle} \otimes 1 \otimes X^{\otimes |b\rangle}] \Delta_{n_b}^{n_a}(V) [X^{\otimes | \neg a \rangle} \otimes 1 \otimes X^{\otimes |b\rangle}] \\ &\in \mathfrak{F}[X, \Delta_{n_b}^{n_a}(V)].\end{aligned}$$

$$\underline{\Delta_{|b\rangle}^{|a\rangle}(V)} \stackrel{\text{Lemma 5.17}}{\in} \mathfrak{F}[X, \Delta_{n_b}^{n_a}(V)] \stackrel{\text{Lemma 5.15}}{\subset} \mathfrak{F}[X, \Delta^1(x), \Delta_1(x), \Delta^{n_a+n_b}(V)]$$

$$\stackrel{\text{Lemma 5.12}}{\subset} \mathfrak{F}[\underline{M, D_q, D_z, \Delta^1(x)}]$$

↑

Def 5.18. $n, x, y \in \mathbb{N}_0$ with $0 \leq x < y < 2^n$

* $V \in U(1H)$, $V = \begin{bmatrix} V_{00} & V_{01} \\ V_{10} & V_{11} \end{bmatrix} \in \{ |0\rangle \langle 0|, |1\rangle \langle 1| \}$.

$$T_{|a\rangle|u\rangle}(V) : {}^1H^{\otimes n} \longrightarrow {}^1H^{\otimes n}$$

$$\sum_{z=0}^{2^n-1} |z\rangle\langle z| + V_{00}|x\rangle\langle x| + V_{01}|x\rangle\langle y| + \underline{V_{10}|y\rangle\langle x| + V_{11}|y\rangle\langle y|}$$

[illegible]

NOTE. $T_{|a\rangle|b\rangle}(V) T_{|a\rangle|b\rangle}(W) = T_{|a\rangle|b\rangle}(VW)$

$$T_{|x\rangle|y\rangle}(V)^* = T_{|x\rangle|y\rangle}(V^*)$$

$$\Rightarrow T_{|\alpha\rangle|\gamma\rangle}(V) T_{|\alpha\rangle|\gamma\rangle}(V)^* = T_{|\alpha\rangle|\gamma\rangle}(VV^*) = 1^{\otimes n}$$

$$\Rightarrow T_{|x\rangle|y\rangle}(V) \in U(H^{\otimes n})$$

Thm 5.20. $n \in \mathbb{N}$, $U \in \mathcal{U}(\mathbb{H}^{\otimes n})$.

Then, $\exists \omega^{(k, k-j)} \in \mathcal{U}(\mathbb{H})$, $k=1, \dots, 2^n-1$, $j=1, \dots, k$.

$$\text{s.t. } U = \prod_{k=1}^{2^n-1} \left[\prod_{j=1}^k T_{|k\rangle|k\rangle}(\omega^{(k, k-j)}) \right]$$

$$\Rightarrow \in \mathcal{F}[T_{|a\rangle|b\rangle}(V)] \text{ for a suitable chosen } V.$$


 $\Delta_{|b\rangle}^{(|a\rangle)}(\omega)$

Thm 5.26.

Goal: $\{M, Dq, D\hat{z}, \Delta'(x)\}$ is universal.

$\Leftrightarrow$ For any $U \in U(\mathbb{H}^{\otimes n})$, $U \in \mathcal{F}[M, Dq, D\hat{z}, \Delta'(x)]$

① $U \in \mathcal{F}[T_{|x\rangle|y\rangle}(V)]$ for a unitary chosen $V \in U(H)$

② $\Delta_{|b\rangle}^{|\alpha\rangle}(V) \in \mathcal{F}[M, Dq, D\hat{z}, \Delta'(x)]$

WTS: $T_{|x\rangle|y\rangle}(V) \in \mathcal{F}[\Delta_{|b\rangle}^{|\alpha\rangle}(V)]$

Def 5.22. $n \in \mathbb{N}$, $x, y \in \mathbb{N}_0$, $0 \leq x < y < 2^n$.

$|x\rangle, |y\rangle$: the computational basis of $\mathbb{H}^{\otimes n}$.

1) Gray-coded transition from $|x\rangle$ to $|y\rangle$:

a finite seq of the computational basis $|g^0\rangle, \dots, |g^{k+1}\rangle$

satisfying the following properties:

① $|g^0\rangle = |x\rangle$, $|g^{k+1}\rangle = |y\rangle$.

② $\forall j \in \{1, \dots, k+1\}$, $\exists n_{a^j}, n_{b^j} \in \mathbb{N}_0$ with $n_{a^j} + n_{b^j} + 1 = n$
 $n_{b^j} \neq n_{b^j}, \forall j \neq j$.

$$\begin{aligned} \text{s.t. } |g^j\rangle &= \underbrace{1^{\otimes n_{a^j}}}_{\text{X}} \otimes \underbrace{X}_{\downarrow} \otimes \underbrace{1^{\otimes n_{b^j}}}_{\text{X}} |g^{j-1}\rangle \\ &= \underline{|(g^{j-1})_{n_{a^j}} \dots (g^{j-1})_{n_{a^j}+1} \neg (g^{j-1})_{n_{b^j}} (g^{j-1})_{n_{b^j}+1} \dots (g^{j-1})_0\rangle}. \end{aligned}$$

$$\text{and } (g^k)_{n_{b^{k+1}}} = 0, \quad (g^{k+1})_{n_{b^{k+1}}} = 1. \quad (*)$$

$$2) |a^{\mathcal{U}}\rangle := |(g^{\mathcal{U}+})_{n_H} \dots (g^{\mathcal{U}+})_{n_H+1}^{\mathcal{U}}\rangle \in {}^1\mathbb{H}^{\otimes n_{a^{\mathcal{U}}}}$$

$$|b^{\mathcal{U}}\rangle := |(g^{\mathcal{U}+})_{n_{b^{\mathcal{U}}-1}} \dots (g^{\mathcal{U}+})_0\rangle \in {}^1\mathbb{H}^{\otimes n_{b^{\mathcal{U}}}}$$

Remark

1. A gray-coded transition from $|b\rangle$ to $|a\rangle$ always exists, but is not unique.

$$2. |g^{\mathcal{U}+}\rangle \neq |g^{\mathcal{U}}\rangle.$$

3. ^{we need} (*) to construct $T_{|a\rangle|b\rangle}^{\mathcal{U}}(V)$ from $\Delta_{|b^{\mathcal{U}}\rangle}^{|a^{\mathcal{U}}\rangle}(V)$.

Thm 5.24.

With the same assumptions in Def 5.22, let $V \in \mathcal{U}(\mathbb{H})$.

Then, every gray-coded transition $|g^{\mathcal{U}}\rangle$, $\mathcal{U} \in \{0, 1, \dots, K+1\}$

from $|x\rangle$ to $|y\rangle$ satisfies

$$\Delta_{|b^{\mathcal{U}}\rangle}^{|a^{\mathcal{U}}\rangle}(x) = \sum_{\substack{z=0 \\ z \neq g^{\mathcal{U}+}, g^{\mathcal{U}}}}^{2^{n-1}} + |g^{\mathcal{U}+}\rangle \langle g^{\mathcal{U}}| + |g^{\mathcal{U}}\rangle \langle g^{\mathcal{U}+}|.$$

$$\cdot T_{|g^{\mathcal{U}}\rangle|y\rangle}(V) = \Delta_{|b^{K+1}\rangle}^{|a^{K+1}\rangle}(V)$$

$$\cdot T_{|g^{\mathcal{U}+}\rangle|y\rangle}(V) = \Delta_{|b^{\mathcal{U}}\rangle}^{|a^{\mathcal{U}}\rangle}(x) T_{|g^{\mathcal{U}}\rangle|y\rangle}(V) \Delta_{|b^{\mathcal{U}}\rangle}^{|a^{\mathcal{U}}\rangle}(x)$$

$$T_{|x\rangle|y\rangle}(V) = \left[\prod_{\mathcal{U}=1}^K \Delta_{|b^{\mathcal{U}}\rangle}^{|a^{\mathcal{U}}\rangle}(x) \right] \Delta_{|b^{K+1}\rangle}^{|a^{K+1}\rangle}(V) \left[\prod_{\mathcal{U}=1}^K \Delta_{|b^{\mathcal{U}}\rangle}^{|a^{\mathcal{U}}\rangle}(x) \right].$$

$$\Rightarrow T_{|x\rangle|y\rangle}(V) \in \mathfrak{H}[\Delta_{|b\rangle}^{|a\rangle}(V)].$$